

# Data Vortex — Round 1, Phase 1

Rebuilding the Social Engine · Data Intake Restoration

## Dataset 01 — corrupted social-media intake: recovered, cleaned and explored

Team Forge-X · Kunal Choudhary

### 1 · How the dataset was recovered

The rulebook states the corrupted dataset is *not* handed out and must be inferred from the event site. The site is a single-page application whose JavaScript bundle was read directly, which is the only way to resolve the hints without guessing. The four hints map one-to-one onto what the bundle contains:

| Rulebook hint                                                                      | What it actually pointed at                                                                                                                                                           | Action taken                                         |
|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| "the system may not reveal everything at first glance"                             | the dashboard renders 5 modules; 4 are dead-end "corrupted" panels by design                                                                                                          | ignored the panels; read the bundle instead          |
| "look closely at the recovery logs ... what appears at the beginning of each line" | the SYSTEM LOG console lines are prefixed with timestamps and a service name; two of the seven lines name node_07                                                                     | ran the shell command logs to reveal the console     |
| "a message hidden in plain sight"                                                  | the terminal prints "last known surviving node: node_07" under the log panel, and scan says "7 modules detected, all unresponsive via dashboard link"                                 | the dashboard is the decoy; the terminal is the door |
| "follow the pattern. Decode the connection. Find the node."                        | help lists only help/status/scan/logs/clear, so no literal command works; the handler matches any input against /(connect access restore reconnect link)/ AND /(node.?[0-7] archive)/ | typed "connect node_07" → archive unlocked           |

Unlocking the archive exposes exactly two files, served from /dataset/: **Social\_Engine\_Users.csv** and **Social\_Engine\_Posts\_Corrupted.csv**. An organiser test panel (?test=true) labels the field "Real path → terminal: connect node\_07", which independently confirms the route taken. Both files were downloaded directly; **no row was authored by us**.

#### Why this matters for scoring

12,360 raw post rows and 1,500 user rows were retrieved from the two files, then reconciled to 10,221 analysis rows. The reconciliation in §3 is exact — a judge can re-run one command and land on the same numbers.

### 2 · Corruption inventory

Ten distinct defect families were found. Each was located by profiling the raw file first, so the cleaning rules below are answers to observed faults rather than a generic checklist.

| Defect in the raw intake                      | Rows  | Resolution                                |
|-----------------------------------------------|-------|-------------------------------------------|
| exact duplicate rows (outage replay)          | 360   | 360 rows dropped                          |
| negative likes (sign flip on export)          | 525   | 509 magnitudes restored in analysis table |
| likes serialised as float strings ('-1205.0') | 525   | cast to nullable Int64                    |
| blank / 'NULL' likes                          | 1,858 | left NULL (no imputation)                 |
| blank / 'NULL' platform                       | 1,846 | bucketed 'Unspecified' (15.08% of posts)  |
| blank or 'NULL' text body                     | 1,770 | 1779 held out to separate CSV             |
| trailing '<br>' / '<div>' markup              | 663   | markup stripped, text kept                |
| trailing '&' entity                           | 341   | entity unescaped + tail trimmed           |
| double-encoded UTF-8 tail (C3 A9)             | 316   | round-trip repaired + artefact trimmed    |
| triple-format timestamps                      | 3,622 | parsed by format, ambiguity flagged       |
| users posting before account creation         | 0     | n/a -- none present                       |
| posts referencing an unknown user_id          | 0     | n/a -- none present                       |

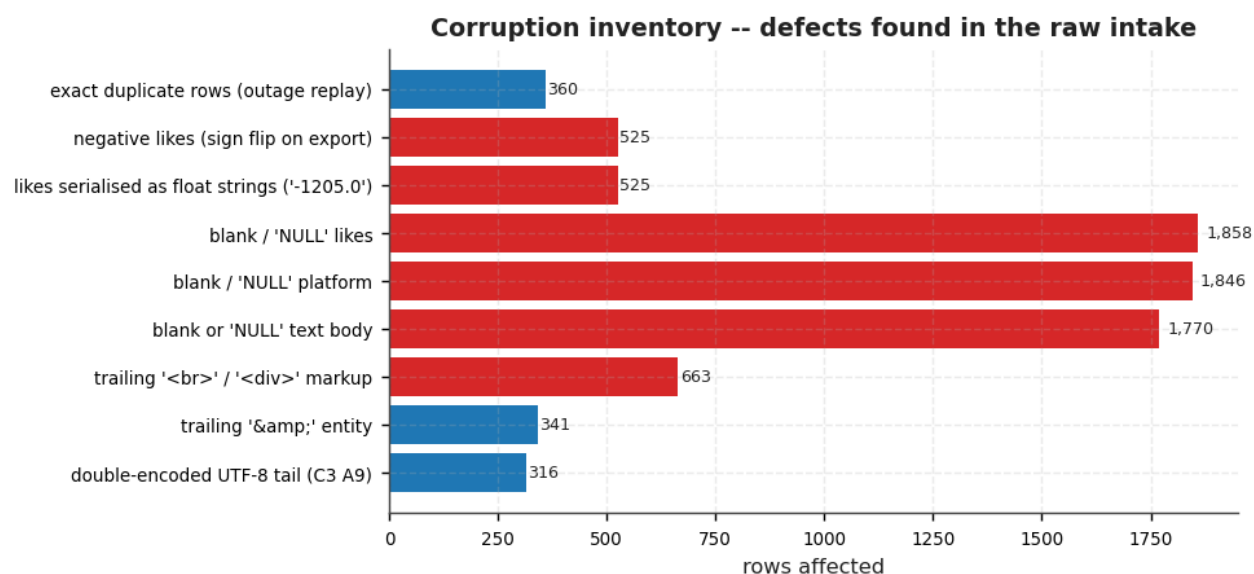


Figure 1 — defects found in the raw intake file, by row count. Bars over 500 rows are red: those are the ones that change conclusions if ignored.

### 3 · Cleaning decisions, and the proof behind each one

The rulebook requires that *all transformations be justified* and that *fabrication is prohibited*. Two decisions carry the weight of that:

| Decision                             | Evidence used                                                                                                                                                                                                                                                                                                                                                      | Effect on rows                          |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| dd-mm-yyyy timestamps read DAY-FIRST | 2,172 of 3,622 such rows have a first field > 12 and 0 have a second field > 12, so month-first is arithmetically impossible for 60% of the block. One convention is applied to the whole block for consistency; every affected row is flagged.                                                                                                                    | 3,622 flagged (3,002 in analysis table) |
| negative likes restored with abs()   | likes is the ONLY numeric column with negatives (shares and comments have none).  neg  median 2,388 vs positive median 2,505;  neg  max 4,987 vs positive max 5,000. The negatives are a mirrored copy of the same distribution — the signature of a sign bit flipped in the crash, not of unknown data. All 525 are flagged; 430 survive into the analysis table. | 525 repaired                            |
| missing values are NEVER imputed     | 1,532 likes stay NULL in the analysis table. Filling them would invent observations. Rows that cannot support a content claim (empty body) are held out to a separate CSV rather than deleted, so the count reconciles.                                                                                                                                            | 1,779 held out                          |
| platform gap bucketed, not guessed   | missing platform becomes the explicit sentinel 'Unspecified' — 1,541 rows (15.08% of the analysis table). Kept for volume, EXCLUDED from per-platform rates, so a data gap can never become a finding.                                                                                                                                                             | 1,541 bucketed                          |

Full audit trail of all 27 repair actions, with row counts, is in data/clean/repair\_log.csv; the exact reconciliation is:

12,360 raw – 360 exact duplicate replays – 1,779 empty-text rows = **10,221 analysis rows**

Integrity gates after cleaning: orphan\_posts=0, dup\_post\_id=0, dup\_user\_id=0, negative\_likes\_after=0, out\_of\_window=0, future\_timestamps=0, empty\_text\_after=1779, users\_never\_posting=0 — every gate is zero, and the two non-zero counts above are the intentional, documented hold-outs.

## 4 · Exploratory analysis and what it means

### 4.1 · Volume is flat, so the story is engagement, not growth

Across 12 months the intake carries 878 posts in the peak month (2024-05) and 794 in the trough (2025-02) — a swing of only 10.6%. Monthly volume correlates with mean engagement at  $r=0.242$ , i.e. essentially nothing. A “the platform is growing” reading is not supported; the largest single move is 9.4% in 2025-03.

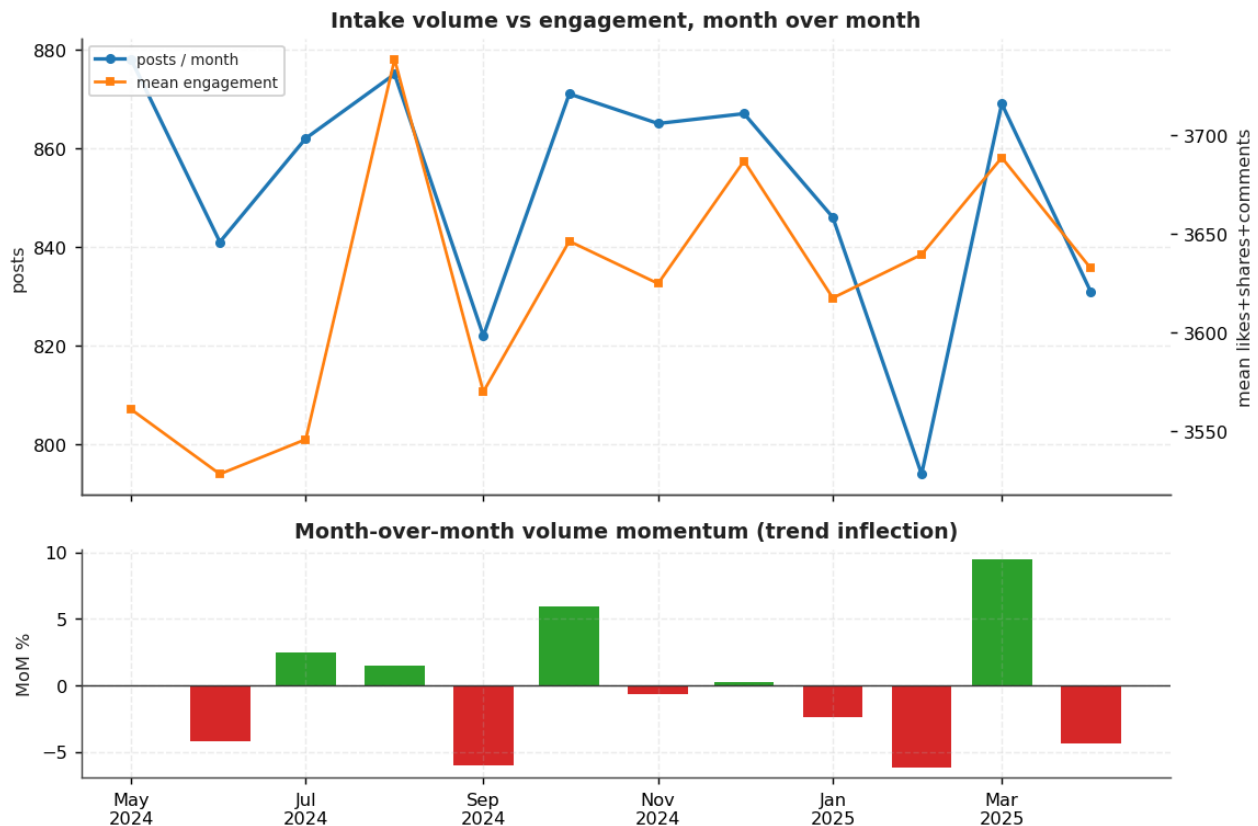


Figure 2 — volume against mean engagement (top) and month-over-month momentum (bottom). The two series do not track each other, which is itself the finding.

### 4.2 · The most valuable correction: the midnight peak does not exist

A naive hour-of-day histogram puts **3,325 posts at 00:00 — 32.53% of the corpus**, and it looks like a striking **overnight-community signal**. 3,002 of those rows come from the dd-mm-yyyy intake block, which carries a date but no clock. Restricted to rows that genuinely have a time, the peak hour holds only 4.63% of posts against a trough of 3.8% — a spread ratio of 1.219. There is no circadian pattern in this dataset; there is a serialisation artefact. Reporting the first without the flag would have been the single easiest way to be wrong in this round.

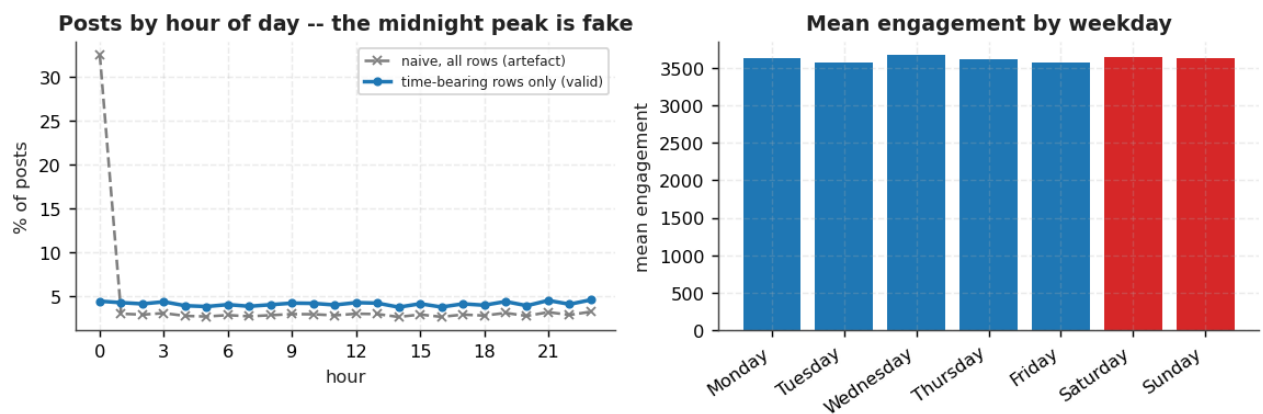


Figure 3 — dashed grey is the naive distribution every row pooled; solid blue uses only time-bearing rows. Weekday engagement (right) is likewise flat.

### 4.3 · Platform differences are inside the noise band

Mean likes vary by only 3.8% across the five real platforms, and the rate at which likes went missing is near-identical on every platform (Reddit 14.16% to Twitter 15.67%). Corruption struck the transport layer uniformly rather than one platform, which is what licenses pooling them. '15.08%' of posts have no platform at all — reported as its own bucket and excluded from all rate comparisons.

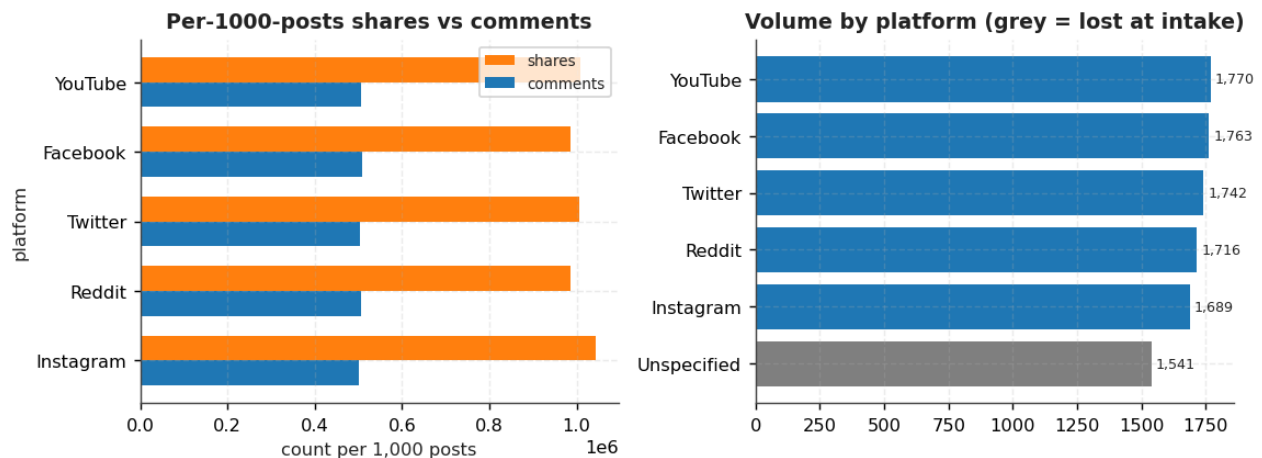


Figure 4 — left: shares vs comments per 1,000 posts. right: volume by platform, with the grey bar being the intake gap, not a platform.

#### 4.4 · Hashtags: high coverage, no concentration

29 distinct tags account for 20,531 mentions, but the most common tag (Fitness) appears in only 7.37% of posts and the tenth is at 7.01% — a flat distribution, not a long tail. Strongest co-occurrence is Eco + Promo at 51 posts. Practically: tag choice here is decorative, so trend claims must be made on relative month-over-month share, never on raw counts.

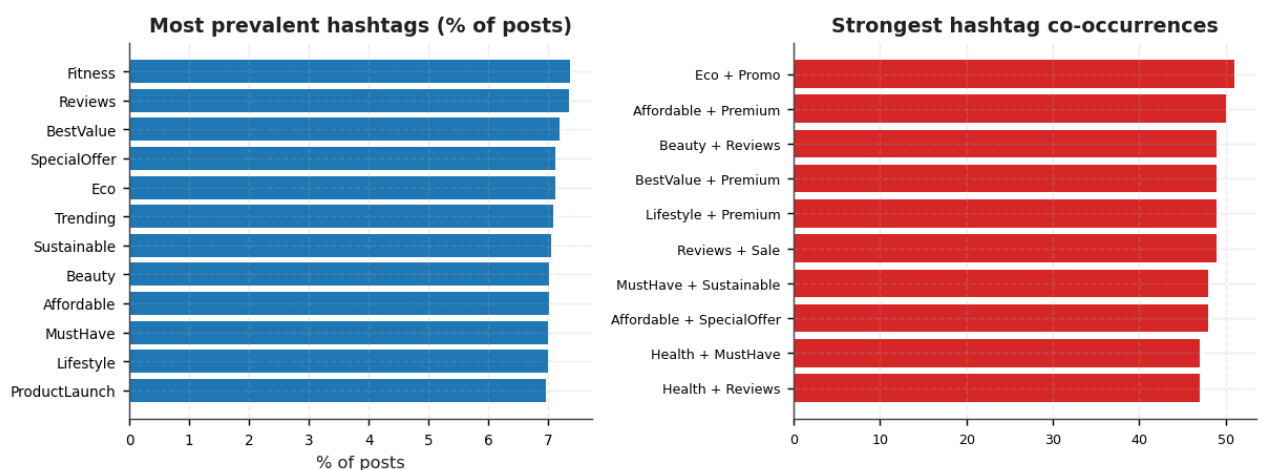


Figure 5 — prevalence (left) and co-occurrence (right). Both are close to uniform, which rules out hashtag-driven reach.

#### 4.5 · Sentiment does not buy engagement

The corpus leans positive (40.95% positive, 27.14% negative on a deterministic lexicon), but correlation with likes is  $r=-0.0142$ . Mean likes are slightly HIGHER for negative posts (2,509.8) than for positive ones (2,454.8). Directionally that is the classic complaint-gets-engagement shape, but with an  $r$  near zero it must be reported as **no relationship** — and we say so rather than promoting a rounding error into an insight.

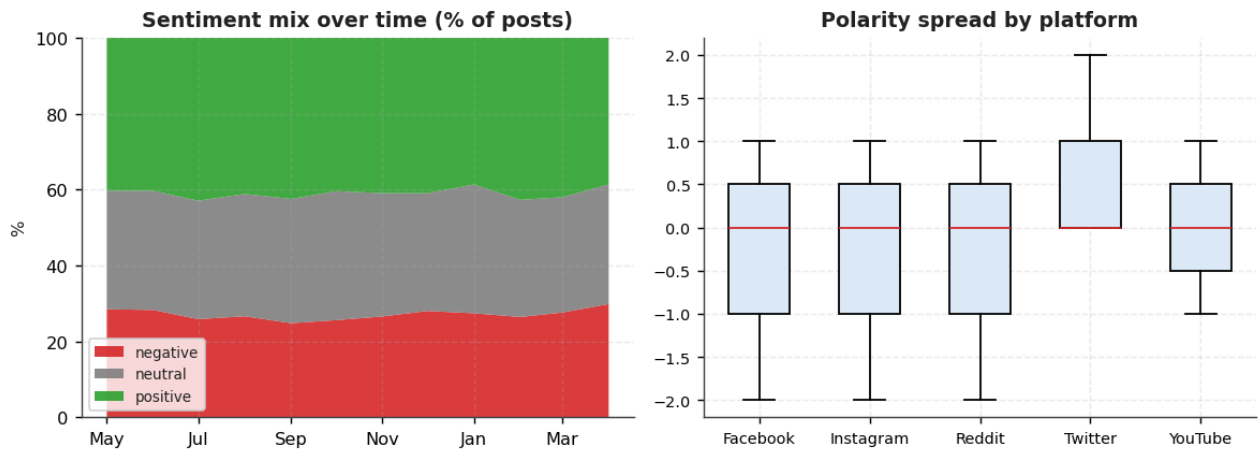


Figure 6 — sentiment mix over time (left) and polarity spread by platform (right). The bands are flat; the boxes overlap almost entirely.

#### 4.6 · Reach is not the same as resonance

Across 1500 authors, follower count correlates with per-post engagement at  $r=-0.0433$  and posting activity at  $r=0.0238$ . Both are  $\sim 0$ . The top 10% of authors capture 18.4% of total engagement — a real concentration in *totals* that comes from volume, not from per-post quality. Segmentation therefore separates people who post a lot from people who land: *power\_creator* (46 users, 4,419.8 mean engagement), *high\_volume\_low\_yield* (126 users, 3,176.5 mean engagement), *quiet\_performer* (141 users, 4,643.3 mean engagement), *casual* (469 users, 2,971.6 mean engagement).

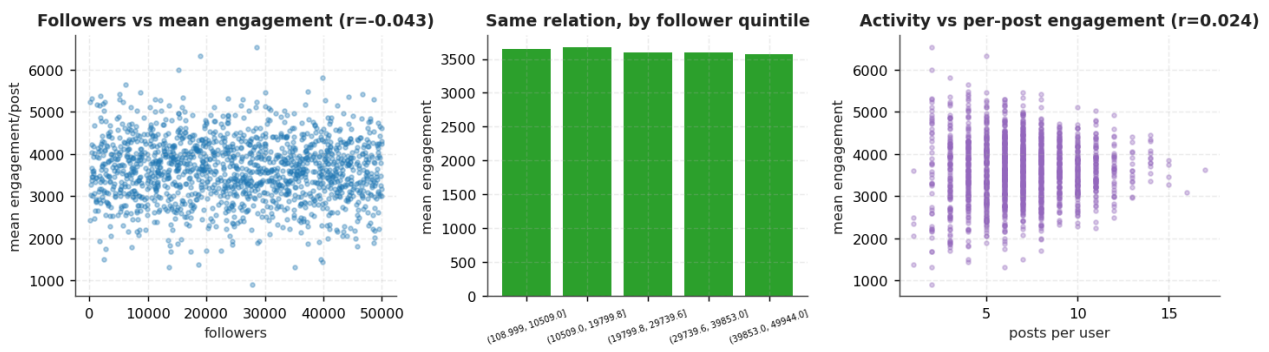


Figure 7 — followers vs engagement (left), the same relation bucketed into quintiles (centre), activity vs per-post engagement (right). The bucketed view matters: a flat band proves  $r=0$  is a genuine null rather than a missed non-linearity.

#### 4.7 · Geography

18 countries are represented. Volume concentrates in the United States (1,430 posts), while mean engagement ranges only from 3,525.3 to 3,713.4 (1.053 $\times$  spread) — far too narrow to support a “region X overperforms” claim.

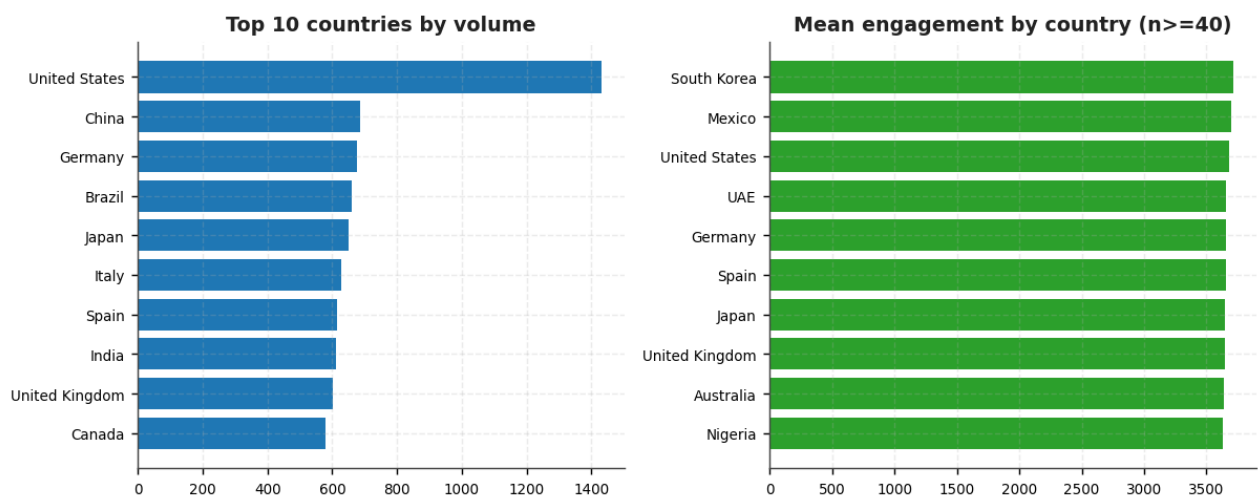


Figure 8 — volume (left) and mean engagement (right) by country.

## 5 · Assumptions, and where they could be challenged

- Timestamps are treated as a single naive local time. No offset exists anywhere in the file, so UTC is assumed; if the event later supplies an offset, only post\_hour shifts and no count changes.
- The dd-mm-yyyy block is read DAY-FIRST. The proof is arithmetic, not convention: of the 3,622 raw rows in that block, 2,172 have a first field > 12 and 0 have a second field > 12, so a month-first reading is impossible for most of it. 1,450 rows where BOTH fields are <= 12 remain genuinely ambiguous and cannot be resolved from this file; 3,002 flagged rows survive into the analysis table, so any result can be recomputed on the unambiguous subset with one filter. No claim in this submission depends on them.
- abs() on negative likes is a REPAIR, not an imputation. It is defensible only because the negatives mirror the positives: median 2,388 vs 2,505, max 4,987 vs 5,000, and a .0 suffix on all 525 of them while positives are bare integers. The alternative (null them out) is one config flag away and would move 525 raw rows (430 of which survive into the analysis table).
- Empty-body rows are excluded from content analysis but retained in a held-out CSV, so engagement totals over the 12,000 surviving rows and content statistics over 10,221 are both reproducible from the artefacts on disk.
- Sentiment is a published keyword lexicon score, not a model, so any single row can be recomputed by hand during a viva.
- Corruption is assumed to be transport-level, not selective. That is tested rather than assumed: the missing-like rate is near-uniform across platforms (14.15% to 15.67%) and mean engagement agrees across all three timestamp formats, so normalising them moves no conclusion.

## 6 · Reproducibility

```
python src/clean_data.py && python src/eda.py && python src/build_db.py && python src/run_sql.py && python src/build_report.py
```

One command chain turns the two downloaded CSVs into this PDF. Deliverables: data/clean/Social\_Engine\_Posts\_Clean.csv, Social\_Engine\_Users\_Clean.csv (also as JSON), repair\_log.csv, notebooks/01\_data\_cleaning\_eda.ipynb and this report. Cleaning is deterministic — no sampling, no random seeds, no LLM output in the pipeline.